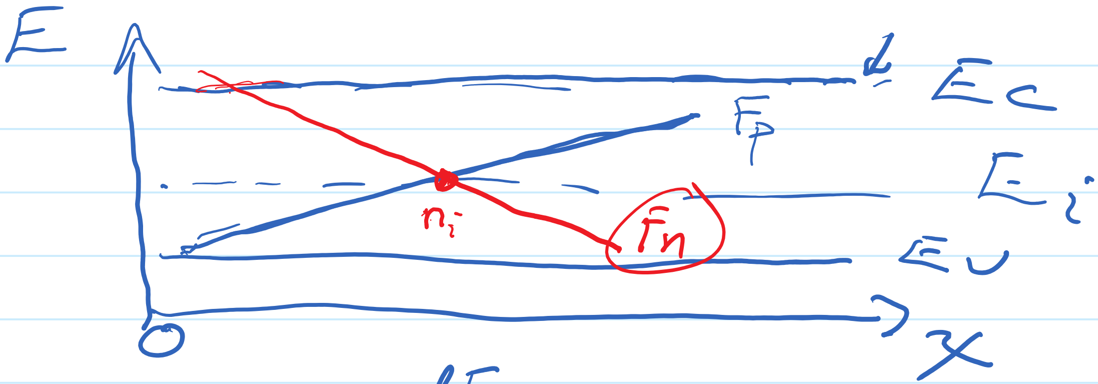



$$J_p = \mu_p p \frac{dF_p}{dx}$$

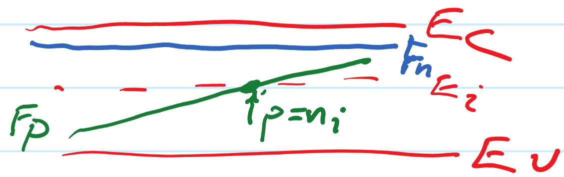
$$J_{p,drift} = \underbrace{q \mu_p p}_{\sigma_p} \mathcal{E}$$

$$\mathcal{E} = \frac{1}{q} \frac{dE_c}{dx}$$

$$\frac{dE_i}{dx}$$

$$\frac{dE_v}{dx}$$

$n = p = n_i$

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$p^+ n$ diode  Double I.

$N_A$  $N_D$

a) double $N_A$

b) 1/2 $N_A$

c) double $N_D$

d) half $N_D$

$$L = \sqrt{D\tau}$$

$$I_0 = q n_i^2 \left( \cancel{\frac{(D_p)}{\underbrace{N_A}_{\approx 100 N_D} L_p}} + \frac{(D_n)}{N_D L_n} \right)$$

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pn jn $V_{bi} = 0.7$ V. $C_{jo} = 2pF$

$V_A$? $\longrightarrow$ $C_j = \frac{C_{jo}}{2}$

$$C_j = \frac{C_{jo}}{\left(1 - \frac{V_A}{V_{bi}}\right)^{1/2}}$$

---

Diode pn jn $\underline{T_{max}\ 100°C - T_{min} = -20°C}$

$V_A = 0.7$ V $\frac{I_{max}}{I_{min}}$.

$\uparrow T$, $I \uparrow \downarrow$

$$I = I_o \left(\boxed{e^{qV_A/kT}} - \cancel{1}\right)$$

$$n_i^2 = e^{-E_G/kT}$$

$$I \propto T^3 e^{-(E_G - V_A)/kT}$$

$$n_i^2 = \underbrace{N_c N_v}_{T^3} e^{-E_G/kT}$$

$$\left(\frac{T_{max}}{T_{min}}\right)^3 \boxed{e^{-(E_G - V_A)\left[\frac{1}{kT_{max}} - \frac{1}{kT_{min}}\right]}}$$

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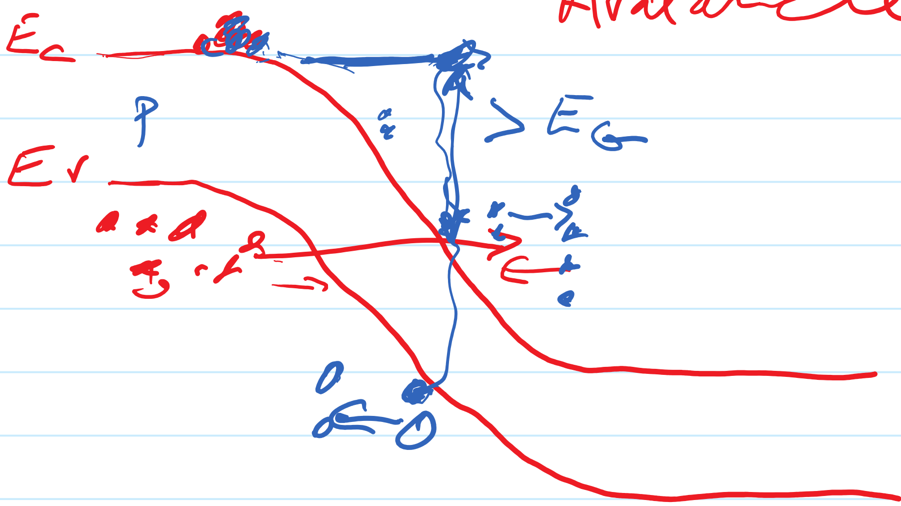
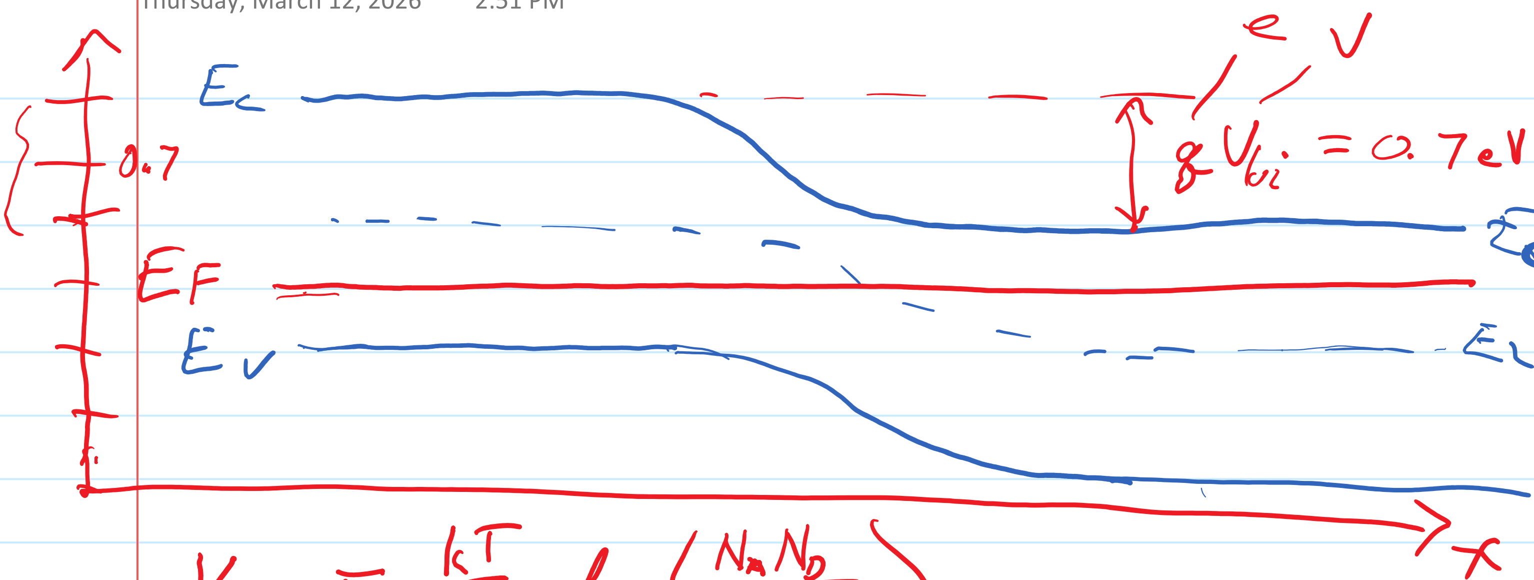
Zener

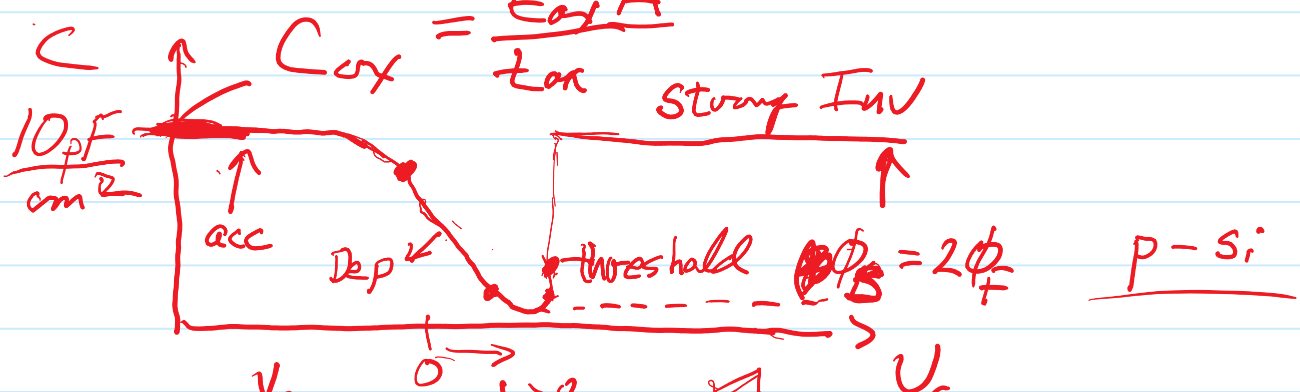
Avalanche

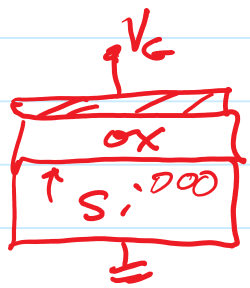


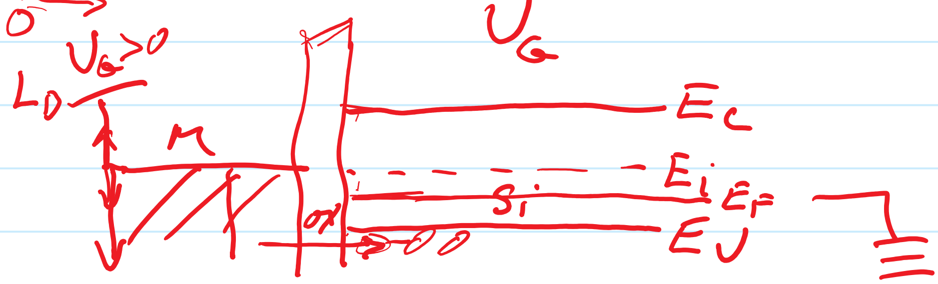



$$V_{bi} = \frac{kT}{q} \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

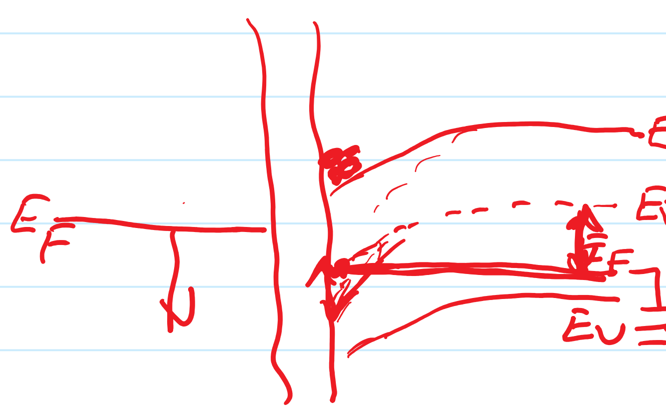
$$C_{ox} = \frac{\epsilon_{ox} A}{t_{ox}}$$





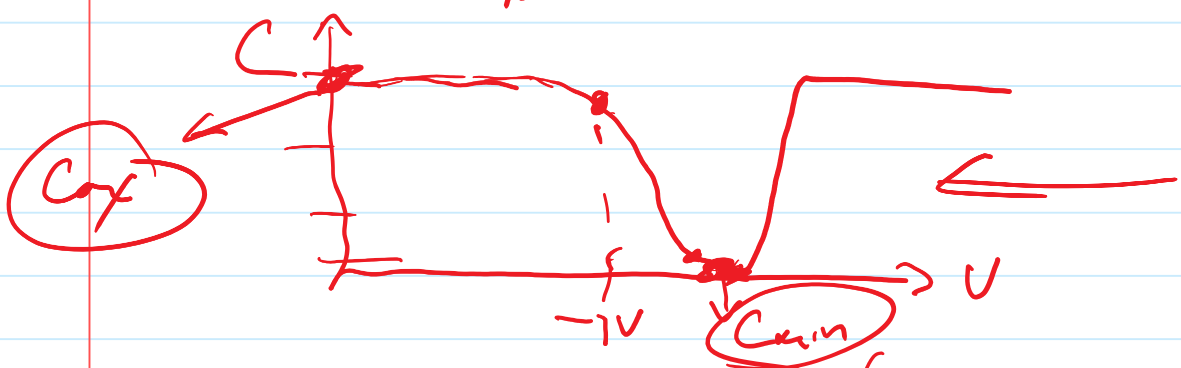


$$\underline{V_t} = V_{FB} + \underbrace{2\phi_F}_{\Delta V_{Si}} - \underbrace{\frac{Q_D}{C_{ox}}}_{\Delta V_{ox}}$$

$Q_D < 0$

$Q_D = -qN_A W$

$$V_{FB} = \frac{-Q_{SS}}{C_{ox}} + \phi_{ms}$$

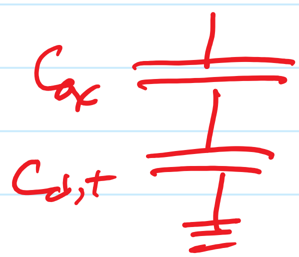



Min. C @ $V_t$ ($C_{ox}$ series $C_{D,t}$)

$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$ $\qquad$ $C_{D,t} = \frac{\epsilon_{Si}}{W_{D,t}}$

$C_{D,t}$ ?

$$\boxed{C_{min} = \frac{C_{ox} C_{D,t}}{C_{ox} + C_{D,t}}}$$




$S = \underline{100}$ mV/dec

$$\frac{I_{on}}{I_{off}} = 10^6$$

$$\underline{S = 60\left(1 + \frac{C_{d,t}}{C_{ox}}\right)}$$

$S_{IDEAL} = 60$ mV/dec @ 300K